

04.04 Operations with Complex Numbers

Common problems and solution methods

Rule guide: recognize the task first

Recognize	Apply these rules
Adding or subtracting complex numbers.	Combine real terms and imaginary terms separately.
Multiplying complex numbers.	Distribute and replace every i^2 with -1 .
Dividing by a complex number.	Multiply numerator and denominator by the denominator's conjugate.

Example 1: Add complex numbers

Problem	$(3 + 2i) + (5 - 7i)$
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How to solve it

Combine like parts:

$$(3 + 5) + (2 - 7)i$$

Simplify:

$$= 8 - 5i$$

Example 2: Multiply complex numbers

Problem	$(3 + 2i)(1 - 4i)$
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How to solve it

Distribute:

$$3 - 12i + 2i - 8i^2$$

Use $i^2 = -1$:

$$= 11 - 10i$$

Example 3: Divide complex numbers

Problem	$\frac{4 + 3i}{2 - i}$
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How to solve it

Multiply by the conjugate:

$$\frac{4 + 3i}{2 - i} \cdot \frac{2 + i}{2 + i}$$

Simplify:

$$= \frac{5 + 10i}{5} = 1 + 2i$$